

On the fly calibration and imaging of streamed TART data

Landman Bester

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Abstract

This note develops the calibration and imaging machinery underpinning the proposed transient detector software pipeline; the detection heuristics themselves are derived in an accompanying document. Because those heuristics apply to calibrated, source-subtracted residual images, on-the-fly direction-independent calibration and calibrator subtraction are a necessary precursor. TART is a single-polarisation, zenith-pointing all-sky array, so we track one scalar complex gain per antenna, written as $g_p = \exp(a_p + i\psi_p)$, together with one real, non-negative flux per catalogued calibrator. The slowly varying log-amplitudes a_p , the faster varying phases ψ_p , and the source fluxes A_s are each modelled as independent integrated Wiener processes (IWPs); see the accompanying document for IWP notation and theory. The driving variances are *not* inferred online—they are calibrated once, offline, from the large body of existing data reductions—which avoids the ill-conditioned coupling between hyperparameter inference and a heavy-tailed likelihood. Conditioned on the gains the model visibility is *linear* in the fluxes, so the fluxes are Rao–Blackwellised out in closed form while the gains are updated by a robust iterated extended Kalman filter: an antenna-based Gauss–Newton step in (a, ψ) , *not* a complex-gain StEFCal iteration. Robustness to the inevitably incomplete satellite catalogue comes from a heavy-tailed (Student- t) observation model, realised as iteratively reweighted least squares. The filter emits gain-corrected, model-subtracted residual visibilities, which are imaged onto a HEALPix grid to produce the residual image stream consumed by the detector.

1 Overview of the imaging and calibration problem

The task is to transform the raw uncalibrated visibilities $V_{pq,t}(\mathbf{b})$ into calibrated residual images $I_t(\mathbf{l})$ in real time, where p, q index the antennas, t the time, $\mathbf{b}$ the baseline vector, and $\mathbf{l}$ the direction on the sky. Because real-time streaming is required, calibration and calibrator source flux estimation needs to happen simultaneously (it is only the calibrator positions which are known a-priori and these are not always 100% accurate). The streaming nature of the problem makes Holoscan a natural fit for the software implementation, with CuPy—rather than an autodiff framework such as JAX—as the numerical backend for the inference machinery. We prefer CuPy because the pipeline must eventually run on constrained hardware, where a single, explicitly budgeted GPU memory pool is important, and because the calibration Jacobian is highly structured (Section 3), so automatic differentiation would buy little (Section 8).

We assume a measurement model of the form

$$V_{pq,t}(\mathbf{b}) = \exp(a_p(t) + i\psi_p(t)) \left(\sum_s A_s(t) X_{pq,ts} \right) \exp(a_q(t) - i\psi_q(t)), \quad (1)$$

where the scalar gain log-amplitudes $a_{p/q}$ are slowly varying real functions of time, the gain phases $\psi_{p/q}$ are more quickly varying real functions of time, the calibrator source fluxes A_s are slowly

varying real functions of time, and $X_{pq,ts}$ is the known unit coherencies of source s to the visibility on baseline pq at time t . The TART telescope [7] always points at zenith and sees almost the whole hemisphere it is located in. This makes spherical coordinates the natural choice for the image domain and the calibrator source positions. The residual images should be computed on a Healpix grid.

Note that calibrator sources are mostly satellites in geostationary orbit, which are bright and have well-known positions but whose fluxes can vary on time-scales of minutes to hours. The gain log-amplitudes and phases are driven by independent integrated Wiener processes (IWPs) with different driving noise levels. The calibrator source fluxes are also driven by independent IWPs, but with much smaller driving noise levels than the gain parameters. Since the calibrator satellite catalogue is almost always incomplete, a robust heavy-tailed likelihood is required to prevent outliers from dominating inference. The IWP state-space model and the heavy-tailed likelihood are combined in a robust iterated extended Kalman filter: the gains are updated by an antenna-based Gauss–Newton step in (a, ψ) , while the catalogue fluxes—which enter the model linearly—are Rao–Blackwellised out in closed form. The filter tracks the calibration parameters, subtracts the calibrator sources, and emits a gain-corrected residual visibility stream that is imaged to produce the residual images fed to the detection pipeline.

The remainder of this note formalises the calibration filter and the imaging step. Section 2 sets up the IWP state-space model and fixes the gauge; Section 3 gives the linearised (iterated-EKF) observation update and its Jacobian; Section 4 adds the heavy-tailed likelihood; Section 5 Rao–Blackwellises the source fluxes and describes their warm-up; Section 6 defines the gain-corrected residual visibilities and their projection onto a HEALPix grid; Section 7 covers offline calibration of the driving variances; and Section 8 collects implementation choices and reuse of existing TART software (the `tart-telescope` organisation and T. Molteni’s repositories, especially `DISKO` and `SPOTLESS`).

2 State-space model for the calibration parameters

Let P be the number of antennas and S_t the number of catalogued sources above the horizon at time t . We track the gain log-amplitudes a_p , the gain phases ψ_p ($p = 1, \dots, P$), and the source fluxes A_s ($s = 1, \dots, S_t$). Each scalar parameter is an independent IWP [3]; following the accompanying note we lift a scalar parameter θ and its first q derivatives into a state $\boldsymbol{\theta} = (\theta, \dot{\theta}, \dots, \theta^{(q)})^\top$ with the exact discrete transition

$$\boldsymbol{\theta}_k = \mathbf{A}_\theta(\Delta_k) \boldsymbol{\theta}_{k-1} + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_\theta(\Delta_k)), \quad (2)$$

where $\mathbf{A}_\theta, \mathbf{Q}_\theta$ are the closed-form IWP matrices of the accompanying document and depend on the per-parameter driving variance σ_θ^2 . We default to $q = 1$ (a locally linear, “constant-velocity” model) for every parameter; the order may be raised per group if the offline data warrant it. Stacking the per-parameter blocks gives the full state $\mathbf{x}_k$ and a block-diagonal transition

$$\mathbf{x}_k = \mathbf{F}_k \mathbf{x}_{k-1} + \mathbf{w}_k, \quad \mathbf{F}_k = \text{blkdiag}(\{\mathbf{A}_{a_p}\}, \{\mathbf{A}_{\psi_p}\}, \{\mathbf{A}_{A_s}\}), \quad (3)$$

with $\mathbf{Q}_k$ the matching block-diagonal process covariance, so the predict step is exact, decoupled across parameters, and $O(P + S_t)$ per frame.

Driving variances are fixed. The driving-variance groups— σ_a^2 (slow), σ_ψ^2 (faster), and σ_A^2 (slow, modulo the warm-up of Section 5)—are *not* estimated online. They are calibrated once from the

existing offline reductions (Section 7). This is deliberate: inferring the driving variances through the marginal likelihood while a heavy-tailed likelihood simultaneously reweights the data is poorly conditioned, the two mechanisms competing to explain the same residual.

Identifiability and gauge fixing. The likelihood depends on the gains only through the products $g_p g_q^* = \exp(a_p + a_q) \exp(\imath(\psi_p - \psi_q))$, which leaves two exact gauge freedoms:

- a global phase $\psi_p \mapsto \psi_p + \phi$ cancels in every $g_p g_q^*$; we remove it by pinning a *reference-antenna* phase, $\psi_{\text{ref}} \equiv 0$;
- a global log-amplitude $a_p \mapsto a_p + c$ rescales every visibility by e^{2c} and is exactly degenerate with an overall flux rescaling $A_s \mapsto e^{-2c} A_s$. Because the catalogue supplies positions but *no* fluxes, this degeneracy cannot be anchored on the source side, so we impose the zero-sum constraint $\sum_p a_p = 0$.

Both constraints are enforced inside the update (Section 3): the reference phase is eliminated from the state, and the gain mean is projected onto $\sum_p a_p = 0$ after each Gauss–Newton step.

3 Observation model and its linearisation

At time t_k each baseline pq contributes a complex visibility. With the single-polarisation scalar gains and the catalogue coherencies $X_{pq,s}$ —unit-amplitude fringes computed from the known source directions (Section 6)—the noise-free model is

$$\hat{V}_{pq} = g_p \hat{M}_{pq} g_q^*, \quad \hat{M}_{pq} = \sum_s A_s X_{pq,s}, \quad g_p = \exp(a_p + \imath\psi_p). \quad (4)$$

We observe $V_{pq} = \hat{V}_{pq} + n_{pq}$ with circular complex noise n_{pq} whose variance is set by the data weights, stacking real and imaginary parts as two real observations per baseline.

Why not StEFCal. StEFCal [4] exploits that, holding all *other* antennas’ complex gains fixed, $\hat{V}_{pq}$ is *linear* in g_p , giving an antenna-by-antenna linear least squares. Our state is (a_p, ψ_p) , not g_p , and $\hat{V}_{pq}$ is nonlinear in these, so we linearise directly (Gauss–Newton / iterated EKF) rather than iterating a complex-gain fixed point. The relevant structure nonetheless survives, and in a cleaner form.

Jacobian. Differentiating (4) at the current estimate,

$$\frac{\partial \hat{V}_{pq}}{\partial a_p} = \frac{\partial \hat{V}_{pq}}{\partial a_q} = \hat{V}_{pq}, \quad \frac{\partial \hat{V}_{pq}}{\partial \psi_p} = +\imath \hat{V}_{pq}, \quad \frac{\partial \hat{V}_{pq}}{\partial \psi_q} = -\imath \hat{V}_{pq}, \quad \frac{\partial \hat{V}_{pq}}{\partial A_s} = g_p X_{pq,s} g_q^*. \quad (5)$$

The gain derivatives are simply the model visibility rotated by 0 or $\frac{\pi}{2}$, so the Jacobian costs nothing beyond $\hat{V}_{pq}$, which we already form. Each baseline row touches only its two antennas: the gain block of the Jacobian inherits the *bipartite* (baseline $\times$ antenna) sparsity of the array’s incidence structure, whereas the flux block $\partial \hat{V}_{pq} / \partial A_s$ is dense (every baseline sees every source). The antenna-based normal-equations structure that StEFCal exploits therefore reappears in $\mathbf{H}_k^* \mathbf{H}_k$, despite the change of parametrisation—which is what makes a hand-coded, structure-aware solver attractive.

Iterated-EKF measurement update. Let $\mathbf{H}_k$ be the stacked (real/imag) Jacobian (5) and $\mathbf{e}_k = V_k - \hat{V}_k(\mathbf{x}_{k|k-1})$ the innovation at the predicted mean. One Gauss–Newton step is the standard EKF update [2]; relinearising about the running estimate and repeating yields the *iterated* EKF, i.e. a Gauss–Newton solver for the MAP state under the IWP prior. A few iterations suffice because the nonlinearity is mild. The normalised innovation (NIS) formed here is logged as the whiteness diagnostic [1] of the accompanying note, but—per Section 2—is never fed back to adapt the driving variances.

4 Robust update via a heavy-tailed likelihood

The Gaussian observation model of Section 3 is fragile: a baseline contaminated by an un-catalogued source or by radio-frequency interference (RFI) is a gross outlier that, under a Gaussian likelihood, would bias every gain and flux. We replace the Gaussian by a Student- t [5] observation density, which has the Gaussian scale-mixture representation

$$t_\nu(\mathbf{e} | \mathbf{0}, \mathbf{R}) = \int_0^\infty \mathcal{N}(\mathbf{e} | \mathbf{0}, \mathbf{R}/\lambda) \text{Gamma}(\lambda | \frac{\nu}{2}, \frac{\nu}{2}) d\lambda, \quad (6)$$

in which the latent per-baseline precision scale λ_{pq} down-weights discrepant data. Treating $\{\lambda_{pq}\}$ by expectation–maximisation (equivalently, one variational step) gives an iteratively reweighted least squares (IRLS) update: at Gauss–Newton iteration j each baseline carries a weight

$$w_{pq}^{(j)} = \frac{\nu + n_y}{\nu + \epsilon_{pq}^{(j)}}, \quad \epsilon_{pq}^{(j)} = \mathbf{e}_{pq}^{(j)\top} \mathbf{R}_{pq}^{-1} \mathbf{e}_{pq}^{(j)}, \quad (7)$$

with $n_y = 2$ for the stacked real/imag residual, and the EKF normal equations are assembled with the reweighted observation precision $w_{pq} \mathbf{R}_{pq}^{-1}$. Inliers keep $w_{pq} \approx 1$; outliers are smoothly suppressed. The degrees-of-freedom ν is fixed (a small value, $\nu \sim 4$, is robust) rather than inferred, again to avoid the coupling of Section 2.

Remark 1 (Two failure modes of an incomplete catalogue). *Robustness here and flux handling in Section 5 address different symptoms of catalogue incompleteness, and both are needed:*

- **Present but un-catalogued** sources have no position, hence no $X_{pq,s}$, and cannot be modelled; they appear as heavy-tailed outliers and are handled only by the Student- t down-weighting.
- **Catalogued but absent** sources do have an $X_{pq,s}$ but no real emission; they are handled by driving their flux state to zero (Section 5).

5 Rao–Blackwellised flux update and warm-up

Conditioned on the gains, the model (4) is *linear* in the fluxes: with g_p, g_q fixed, $\hat{V}_{pq} = \sum_s A_s (g_p X_{pq,s} g_q^*)$. Partition the state into the nonlinear gains $\mathbf{x}_g = \{a_p, \psi_p\}$ and the linear fluxes $\mathbf{x}_A = \{A_s\}$. The joint predicted density is Gaussian (block-diagonal, the IWPs being independent) and the flux-conditional posterior $p(\mathbf{x}_A | \mathbf{x}_g, V_k)$ is exactly Gaussian. We therefore *Rao–Blackwellise* the fluxes: rather than sampling or jointly Newton-stepping them, we eliminate $\mathbf{x}_A$ in closed form via the Schur complement of the (reweighted) normal-equations matrix, so the gain update sees the fluxes marginalised with their full uncertainty and the flux posterior is recovered by back-substitution. This lowers the variance of the gain estimate and confines the per-frame nonlinear solve to the $2P$ -dimensional gain block.

Non-negativity. A physical flux satisfies $A_s \geq 0$. We keep A_s *linear* (not log-parametrised like the gains) so the Rao–Blackwell step stays exact, and impose non-negativity as a constraint on the flux block, turning its update into a small non-negative least squares (a truncated-Gaussian posterior); the Schur elimination then runs on the *active* (strictly positive) set each frame. Non-negativity is what makes a genuinely absent source harmless: lacking any positive likelihood gradient, and forbidden from going negative to absorb noise, its flux estimate simply stays near zero.

Warm-up schedule. At acquisition the fluxes are unknown and the gains are not yet locked, so we initialise every flux at mean zero with a diffuse covariance and run the flux driving variance σ_A^2 *large*, letting the data raise the fluxes of sources that are truly present. As the filter converges—monitored via the settling of the gain-block NIS rather than a fixed frame count— σ_A^2 is annealed down to the realistic floor calibrated offline (Section 7); an exponential schedule is the fallback. Because a large warm-up σ_A^2 also lets noise excite *absent* sources (which then freeze when the variance tightens), the zero-mean initialisation, the non-negativity constraint, and the Student- t weights together bias those fluxes back down. Should residual leakage into absent sources be observed, a mild ℓ_1 (LASSO-style proximal) shrinkage on $\mathbf{x}_A$ supplies *active* suppression; we start without it and add it only if needed.

6 Calibrated residuals and all-sky imaging

Coherencies from the catalogue. TART points at zenith and the satellite catalogue gives positions (from broadcast ephemerides / two-line elements), not fluxes. For source s with time-dependent direction cosines $\hat{\mathbf{l}}_s(t) = (l_s, m_s, n_s)$ in the array frame and baseline $\mathbf{b}_{pq} = (u_{pq}, v_{pq}, w_{pq})$, the unit coherency is the geometric fringe

$$X_{pq,s}(t) = \exp\left(2\pi i \frac{\nu}{c} [u_{pq}l_s + v_{pq}m_s + w_{pq}(n_s - 1)]\right), \quad n_s = \sqrt{1 - l_s^2 - m_s^2}. \quad (8)$$

Satellites in medium Earth orbit move appreciably over minutes, so $X_{pq,s}$ is recomputed every frame from the propagated positions; the motion is in fact helpful, modulating each source’s fringe and so aiding the flux/gain separation. The coordinate pipeline (ephemeris propagation, ECEF→topocentric→direction cosines, horizon masking) is the natural place to reuse TART community code (Section 8).

Gain-corrected residual visibilities. After the update the filter forms the model $\hat{M}_{pq} = \sum_s \hat{A}_s X_{pq,s}$ and emits, per baseline, the gain-corrected, model-subtracted residual together with its corrected weight,

$$R_{pq} = \hat{g}_p^{-1} V_{pq} \hat{g}_q^{-*} - \hat{M}_{pq}, \quad w_{pq}^{\text{corr}} = w_{pq} |\hat{g}_p \hat{g}_q|^2. \quad (9)$$

Working in the gain-corrected (sky) frame puts the residual in physical flux units with a well-characterised noise variance—exactly the per-frame variance R_k the detector consumes. By construction R_{pq} has the same layout as a raw visibility, so the downstream imager need not know that calibration occurred.

HEALPix imaging. The residual map is the adjoint (gridless) DFT of R_{pq} evaluated at the centres of the visible HEALPix pixels [6] rather than on a tangent-plane (l, m) grid:

$$I_t(\hat{\mathbf{l}}_j) = \frac{1}{n_j} \text{Re} \sum_{pq} w_{pq}^{\text{corr}} R_{pq} \exp\left(-2\pi i \frac{\nu}{c} [u_{pq}l_j + v_{pq}m_j + w_{pq}(n_j - 1)]\right), \quad (10)$$

the natural representation for a zenith-pointing instrument that sees most of the hemisphere (a single tangent plane breaks down at large zenith angle). The existing tiled (l, m) DFT operator is a structural template: the per-pixel phase, the w -term $(n_j - 1)$ correction, and the memory tiling all carry over once the pixel coordinates are HEALPix direction cosines. Each frame yields one all-sky residual map; the stream of maps is the per-pixel light-curve input to the detector, whose quiescent per-pixel noise level follows from $\{w_{pq}^{\text{corr}}\}$.

7 Offline calibration of the driving variances

The driving variances are estimated once, offline, from the existing body of data reductions and then frozen for the online filter. Given an offline gain solution sampled at $\{t_k\}$, the IWP increments of each log-amplitude and phase track are (for $q = 1$) approximately $\theta_k^{(q)} - \theta_{k-1}^{(q)} \sim \mathcal{N}(0, \sigma_\theta^2 g(\Delta_k))$, with $g(\Delta)$ the relevant entry of $\mathbf{Q}_\theta(\Delta)$, so σ_θ^2 is recovered by matching the empirical increment variance (equivalently, by maximising the IWP marginal likelihood of the solution track). The same procedure on offline flux solutions sets the post-warm-up floor of σ_A^2 , and the spread of fitted values across observations sets sensible defaults—they change little between observations, which is precisely why fixing them is safe. The fitter is a small offline utility emitting a frozen hyperparameter configuration; it never runs inside the streaming pipeline.

8 Implementation notes

The pipeline is realised as a streaming graph of separate operators—reader $\rightarrow$ sky-model $\rightarrow$ calibration $\rightarrow$ imaging $\rightarrow$ writer—so each stage has a single responsibility and can be tested in isolation:

- **Sky-model operator** computes $X_{pq,s}(t)$ from the catalogue and the per-frame geometry. TART is single-polarisation, so one scalar coherency per source/baseline suffices. The ephemeris/coordinate logic can be borrowed from TART community code (the `tart-telescope` organisation; T. Molteni’s DISKO for HEALPix imaging and SPOTLESS for visibility-domain source removal) [8], but those are host/NumPy implementations—we re-implement the per-frame kernels as GPU (CuPy) operators rather than calling them at runtime.
- **Calibration operator** is stateful (it carries the filter mean/covariance, the gauge, and the warm-up schedule across frames), owns the subtraction, and emits the gain-corrected residuals (9) plus a solutions side-stream (gains, fluxes, NIS) for diagnostics and for the offline hyperparameter fit.
- **Imaging operator** is a new HEALPix DFT operator, structured after the existing tiled (l, m) operator, consuming residuals in the raw-visibility layout.
- **Reader** is deferred: TART data are not delivered as MSv4 Zarr, and the concrete ingest format will be fixed once sample data are in hand. The existing MSv4 reader serves only as a structural template.

The numerical backend is CuPy throughout, which also lets the calibration and imaging operators exchange GPU buffers with the existing CuPy/Holoscan tensors at zero copy. Because the calibration Jacobian is structured (antenna-bipartite gains, dense flux block), the update is hand-coded structured linear algebra rather than relying on a high-level autodiff framework.

References

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